

Putting it all together

DSC 340 Applied Machine Learning

Operationalization [EMC 12]

May 5, 2025

The Plan

- Revisit:
 - Roles in an Analytics Project
 - Data Analytics Lifecycle
 - Developing an Analytics Plan
 - Organizational details
- Reporting and Operationalizing Results
- [if time] Final Project Team/Work Time

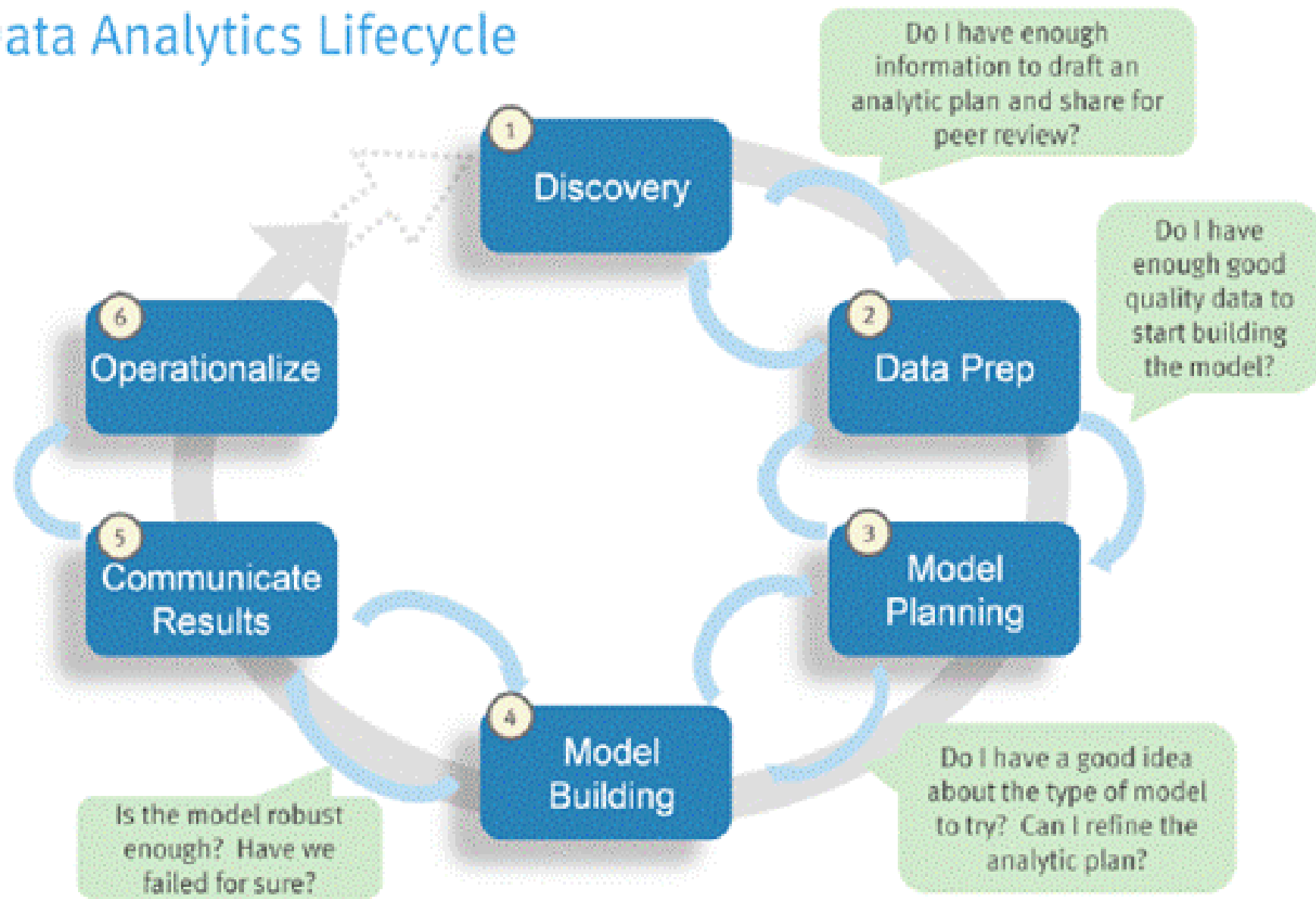
Announcements

- Last Class Days:
 - Wed. May 7 and Fri. May 9: Project Workdays (classroom)
 - Monday, May 12: **Presentations (Lightning Talks)**
 - Tuesday, May 13: Jupyter Notebook (all code + written report) due 11:55 PM
- Comprehensive Final Exam: Friday, May 16th (1:30 - 4:30 PM)
 - Same format as tests
 - About 2x length of tests
 - about 30 MCQs @ 2pts => 60 points
 - about 4 problems/short essays @ 10 pts => 40 points

Data Analytics Roles (Revisited)

- **Organization User:** understands domain area, benefits from work
- **Project Sponsor:** high-level direction from administration
- **Project Manager:** supervises process to meet milestones/objectives
- **BI Analyst:** Business domain expertise, dashboards and reports
- **DB Administrator:** IT support for DB: roles, permissions, installations
- **Data Engineer:** Programming, complex SQL queries, data munging
- **Data Scientist:** The expert in the room for techniques, models, and evaluating result: Choose the right method for the problem at hand.

Data Analytics Lifecycle



Developing an analytics plan

- Interview client: include stakeholders
- Do we understand the problem?
- Do we have access to the data we need?
- Do we have the other resources we need?
 - Computational resources
 - Human resources, skillsets
- Develop timetable with landmarks
- Discuss/negotiate/adapt plan with client
- Keep client in the loop (schedule regular meetings/updates)

Organizational details

- Regular collaboration with team
- Shared calendar
- Shared repository with:
 - Raw data
 - Prepared data
 - Code (scripts, notebooks)
 - Visualizations
 - Reports, presentations, etc.
- Data integrity, security, regular backups

Toy example (as always!)

- Problem: Bocce ball turnout
- Predictive model:
 - How many bocce balls?
 - How many individually wrapped snacks?
 - How many bottled beverages of each type?
- **Operationalize: dashboard** for event organizers
 - **Prescriptive analytics** vs. **predictive analytics**
 - Place on organization's intranet (accessible locally/VPN)
- Larger project: nationwide rollout if pilot at Centre successful 😊

Summary of Operationalization Phase

- Present final reports
- Assess benefits of analytics project
 - Descriptive: What has happened in the past
 - Predictive: Forecasting for the near future
 - Prescriptive: Support optimal decision making
- Set up pilot program to deploy models
- Larger rollout

Stakeholders

- **Business User:** determines benefits and implications of findings
- **Project Sponsor:** business impact, risks, ROI, how to promote project
- **Project Manager:** project completion, deadline, and budget
- **BI Analyst:** how existing reports and dashboards will be impacted
- **Data Engineer and DBA:** give them analytics code + technical docs
- **Data Scientists:** Share code, explain model to peers, managers, etc.

Deliverables

- **Presentation for Project Sponsors**
 - High-level takeaways for executives
 - Key messages to aid decision-making
 - Focus: clean, easy visuals for presenter to explain and viewer to grasp
- **Presentation for Analytics Team**
 - Describe changes to business processes and reports
 - Visuals include ROC curves, density plots, histograms, etc.
- **Code:** for engineers and others in production environment
- **Technical specifications:** for implementing the code

To Report

- **Project Goals**
- **Main Findings:** Summary, emphasize key messages
- **Approach:** High-level methodology
- **Model Description:** Overview of modeling technique
- **Key Points Supported with Data** (e.g., bar chart)
- **Model Details** (omit for sponsor)
- **Recommendations** (different for sponsor and analysts)